The Galaxy Workflow Draft Format

`class: GalaxyWorkflowDraft` · gxformat2 that can say “not yet”, checked by gxwf

Galaxy PROJECT

Make “I don’t know yet” a typed, checkable state. Then burn it down.

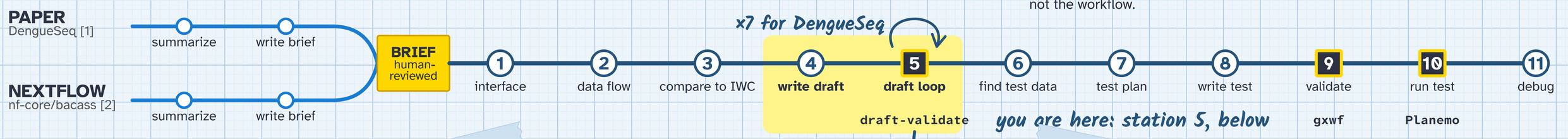
31 → 0

A **Galaxy Workflow Draft** is gxformat2 with named holes. Inputs, outputs, steps and every edge are concrete from the start; tool ids, versions and port names may wait as typed `TODO_*` sentinels, with `_plan_*` notes holding the intent. `gxwf` checks what is concrete, picks what to resolve next, and promotes the finished draft to plain gxformat2. Only the boxes are incomplete, never the wiring.

Where drafts fit

Highlighted, the loop: validate → next step → resolve, then promote.
Two real agent runs on one route, from a Galaxy Workflow Foundry run.

■ a program checks the work here
DengueSeq: test **passed**. bacass: 1 assertion failed; debug blamed the assertion, not the workflow.



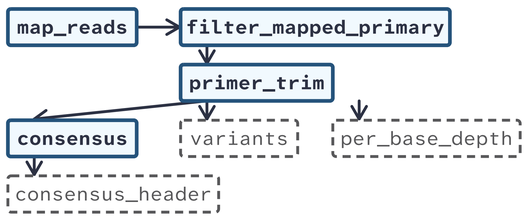
1 Known knowns

Inputs, outputs, steps and every edge are concrete from the first draft.

3 Valid at every step

`draft-validate --concrete` *checks concrete steps against real tool schemas and skips the rest.*

DRAFT AT `daf2928` · 4 of 7 concrete



Solid: checked against the tool’s schema. Dashed: ports still TODO_, skipped. Edges all there. **Valid anyway.**

The format

Always concrete: inputs, outputs, step labels and types, every edge.
May wait: `tool_id`, `tool_version`, port names, as TODO or TODO_<hint>.
Intent: `_plan_state`, `_plan_context`, `_plan_in`, `_plan_out`. Gone once the step resolves.
Promote: `gxwf draft-extract` strips `_plan_*` and flips the class to `GalaxyWorkflow`.

5 Promote, don’t translate

Fully resolved, the draft is ordinary gxformat2. No translation layer.

Get started

```
npm install -g @galaxy-tool-util/cli
gxwf draft-validate \
  --concrete wf.gxwf.yml
gxwf draft-next-step wf.gxwf.yml
```

GALAXY WORKFLOW DRAFT

STATEMENT OF KNOWN UNKNOWNS

station 5 · run dengueseq-denv3-galaxy · worked example

OPENING BALANCE bd40715 ..... **31** TODO

```
class: GalaxyWorkflowDraft
steps:
  map_reads:
    tool_id: …/devteam/bwa/bwa_mem
    tool_version: TODO
    in:
      TODO_paired_reads: Paired-end reads
      TODO_reference_fasta: DENV3 reference genome
    out:
      - id: TODO_coordinate_sorted_bam
    _plan_state: | BWA-MEM paired-end …
  filter_mapped_primary: …  # 6 more steps
```

BURN-DOWN ▸ picked by draft-next-step

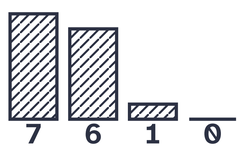
		VALIDATE	TODO LEFT
template	bd40715 · 7 steps planned, 0 concrete	ok □□□□□□□	31
▸ **map_reads**	80de2f3 · bwa_mem 0.7.19+galaxy1	ok ■□□□□□□	27
▸ **filter_mapped_primary**	3556725 · samtools_view 1.22+galaxy2	ok ■■□□□□□	24
▸ **primer_trim**	658af4e · ivar_trim 1.4.4+galaxy1	ok ■■■□□□□	19
▸ **consensus**	daf2928 · ivar_consensus 1.4.4+galaxy0	ok ■■■■□□□	13
▸ **per_base_depth**	d366ab3 · bedtools_genomecoveragebed 2.31.1+galaxy0	ok ■■■■■□□	9
▸ **variants**	07062a2 · ivar_variants 1.4.4+galaxy0	ok ■■■■■■□	4
▸ **consensus_header**	b82fa5e · tp_sed_tool 9.11+galaxy0	ok ■■■■■■■	0

BALANCE DUE . . . . . PROMOTED plain gxformat2 . **0**

b82fa5e writes galaxy-workflow.gxwf.yml
GalaxyWorkflowDraft → **GalaxyWorkflow**

SECOND RUN, SAME LOOP

bacass-sr-assembly-galaxy
from Nextflow · 3 steps resolved
valid at every commit · promoted ad86a85



Two real agent runs: DengueSeq DENV3 [1] (top), nf-core/bacass [2] (stub). Every draft commit replayed with gxwf 1.15.0 draft-validate --concrete --connections. This shows resolution, not correctness.

2 Known unknowns

Versions, ports, anything not yet known: typed TODO_* sentinels. _plan_* fields carry the intent forward.

4 The draft says what’s next

`gxwf draft-next-step` *picks the next step. An agent, or you, finds its tool on the Tool Shed, pins a revision, fills the step. The format drives the loop.*

6 Reviewable intent

Each resolution is one commit: a TODO out, a pinned tool in. Review the diff.
`_plan_*` notes keep the why beside the step and can’t go stale: a resolved step that still carries them fails validation. `draft-next-step` holds no hidden state; the same draft gives the same answer.

7 Unknown unknowns

Those take a run: bacass passed every static check, then failed one test assertion. Checks prove shape, not biology. A validated workflow is a hypothesis about the analysis until it has run.

Read more

Format spec and every `gxwf` flag:

`jmchilton.github.io/galaxy-tool-util-ts`
Guides → Draft Workflow Format

In code: `@galaxy-tool-util/schema` exports validate, next step and extract as functions, plus a JSON Schema for editors.

In use: the Galaxy Workflow Foundry’s agent skills run this loop, from paper to tested workflow.

[1] Vogels CBF, Hill V, et al. DengueSeq. BMC Genomics 25:433 (2024).

[2] nf-core/bacass 2.5.0, github.com/nf-core/bacass